

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Chicago, Illinois 60607

Homework Assignment #11

April 7, 2020

(due 11:59 AM, April 10, 2020)

Do both problems.

An uncharged membrane separates two aqueous salt solutions that contain a protein at concentrations of C_L and C_R , with $C_L > C_R$ (see Figure *). Stirring the solutions reduces, but does not eliminate, mass transfer effects near the membrane surface. The salt concentrations are the same for both solutions, so the

potential differences are negligible: Figure * shows concentration distributions for the protein solutions.

For each sketch, briefly discuss whether the concentration profiles are physically possible, and if they are, determine whether the partition coefficient $\Phi (= C_m/C_L \text{ or } C_m/C_R)$ is greater than, less than, or equal to unity.

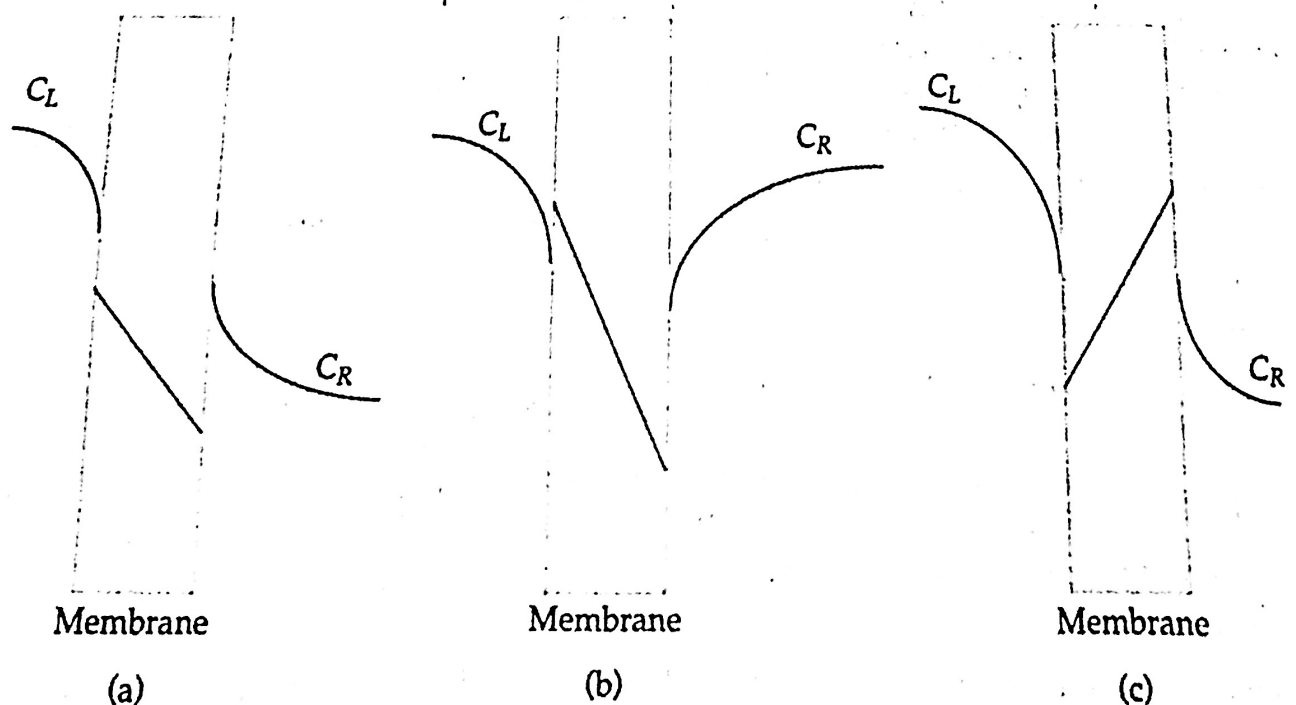


FIGURE * Diffusion across a membrane with a partition coefficient different from 1.

Consider a two-layer model of an artery, as shown in Figure #. The layers are of thickness $R_0 - R_1$ and $R_1 - R_i$. The inner layer has the diffusion coefficient D_i and the outer layer has the diffusion coefficient D_0 . The solute concentration in the lumen (i.e., $0 < r < R_i$) is C_i , and the concentration at R_0 is C_0 . Calculate the effective diffusion coefficient.

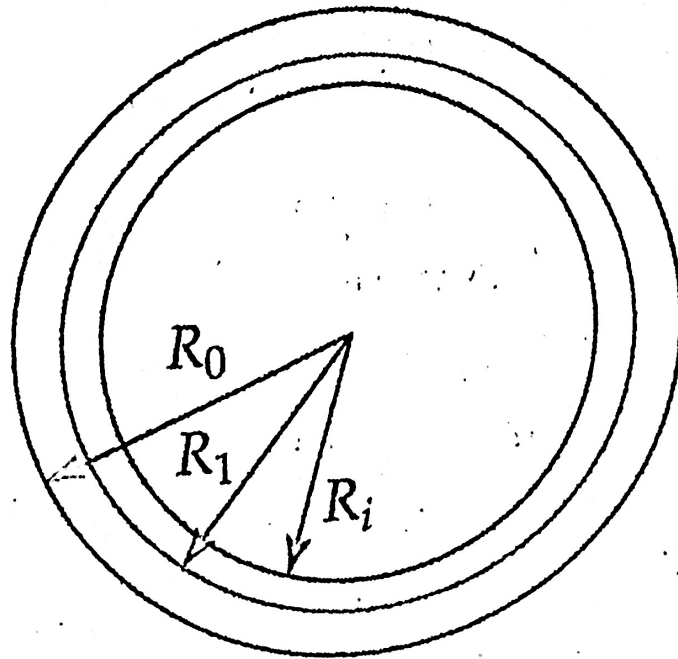


FIGURE # Diffusion through a cylindrical laminate.